Written Assignment

Name:Chen YiChi

UESTC ID:[redacted]

Assignment Title: Challenges in Assisting Elderly Independent Living by Ambient Assisted Living and Methods of Resolving Them

Word Account:1050

Challenges in Assisting Elderly Independent Living by Ambient Assisted Living and Methods of Resolving Them

Abstract

Ambient Assisted Living(AAL) could assist elderly to improve their independency, which becomes increasingly important in the context of an ageing population. However, several issues still exist. Special consideration should be given to factors including manual interaction, convenience of equipment and data security. Otherwise, elderly's acceptance of AAL would decrease due to the unsatisfactory user experience or concerns about data leakage. Radar system could resolve the issues because it does not need to record or be carried around. SafeNeighborhood(SN) is also a promising system which merges information from the whole neighborhood to create a private and safe environment for elderly. But these two methods both have technical flaws. Hence, this report suggests that a third approach, applying IOT in AAL, worth focusing on since this strategy could not only solve the issues of data leakage and redundant manual interaction but also provide helpful assistance.

[143 words]

Ambient Assisted Living (AAL) refers to a collection of facilities aiming to solve problems arising from aging of population [1]. It utilizes technologies based upon ambient intelligence. Nowadays, longevity results in slow physical aging, which poses elderly's increased dependency on others [2]. Ambient assisted living (AAL) extends their capacity to maintain an active and independent lifestyle as they get older [2].

To be more specific, AAL technologies provide an ecosystem of various sensors, wireless networks, mobile devices, computers, and software programs designed for monitoring personal, environmental, and healthcare needs [3]. However, challenges still remain in many aspects, such as acceptance, trust, ethic, privacy and techniques. This report will briefly introduce several issues of promoting AAL to assist elderly

independent living. It will then present three methods which could resolve those issues and give an evaluation to select the ideal option.

According to [4], barriers in acceptance of AAL Technologies are factors that can't be ignored. The majority of elderly stick to their familiar lifestyle and resist any changes in their behaviors or daily routines [4]. They also dislike devices that requires their great effort to perform. Besides, elderly are concerned about the potential misuse of their personal information and are resistant to be monitored by cameras and other technological means [4.5]. These factors create difficulties in promoting the adoption of AAL.

These factors mainly attribute to elderly's cognitive, perceptual, or physical limitations [6]. Therefore, designing AAL systems for the elderly requires special attention in many aspects. In the first place, Ambient Assisted Living (AAL) systems should ideally operate without significant effort from the user [4, 6]. Otherwise, this will add to the burden of these older users and make them more reluctant to use AAL. Secondly, Special consideration must be given to sensors and other equipment during the design of Ambient Assisted Living (AAL) systems. Since sensors usually come into direct contact with the body, their size and compatibility with human tissues deserve careful consideration [5, 6]. Inappropriate design can cause inconvenience to the life of the elderly, which is contrary to our original design intention. Last but not least, privacy and confidentiality of users' data are of vital importance to AAL promoting [6]. This aspect is particularly crucial for wireless communication which is easy to intercept [6]. Uncritical data transmission and collection methods could trigger data tampering and data leakage.

One method is to use radar system to replace other monitors like cameras and wearable sensors [5]. Firstly, radar has superior capacities. It excels in detecting human presence, monitoring their movements within a certain area and detailing these movements, from bulk body motion to tiny movements of specific body parts, and even capturing subtle actions in chest or abdomen during breathing or heartbeats [5]. Moreover, Radar possesses two incomparable advantages against cameras and wearable sensors [5]. Compared with cameras, radar systems do not save visual

images or video recordings of monitored objects, thus easing elderly's concerns to a greater extent [1, 5]. Besides, unlike wearable sensors, radar systems do not require users to wear, carry, or interact with other electronic device, nor do they cause changes to elderly's everyday habits and behaviors., thereby enhancing the technology acceptance [1, 4, 5]. However, it still has some technical deficiencies. One biggest challenge is how to avoid incorrect alerts during fall detection tasks when the user sits or bends down [5].

Another approach is SafeNeighborhood(SN), which is an Ambient Assisted Living (AAL) system integrating data from various sources with collective intelligence to adjust sensor data [2]. It incorporates mobile, ambient, and AI technologies, alongside Old Neighborhood Relationships, to create secure outdoor environments. The developing system enhances the independence of the elderly and does not need their direct and frequent interaction with technology [2], which can largely improve their acceptance of AAL [4]. Furthermore, the system adopts a trusted, private, neighborhood-based social network, only allowing trustworthy individuals to have the access to elderly's information [2, 4]. This initiative effectively protects the privacy of individuals [2]. However, this technology is still under development and has a number of technical shortcomings. Currently, SN only caters to users who have declining senses due to aging [2]. It is not able to tackle severe disabilities like Alzheimer's disease, blindness, or deafness [2].

Both of the methods described above are technically unsophisticated. The final approach I would present in this paragraph deploys a prevalent and mature technology called internet of things (IOT) [7]. It is mainly accomplished by two technologies. The first technology is Keep in Touch(KIT).It is designed to monitor elderly's health status by collecting and transmitting health data [7]. Using mobile phones equipped with KIT NFC, users have the capacities to collect application-specific data just by touching the corresponding object. In the whole system, mobile phones serve as universal terminals for medical devices and smart objects. This technology has two major advantages.

Data can be accessed without manual interaction on the mobile phone. Apart from this, maintenance and management of devices consume lower effort compared with other

methods [6, 7]. The second method applied in IOT is Closed Loop Healthcare Services. It is designed to minimize the insufficiency in traditional central storage of relevant health data [7]. The closed loop consists of three parts. The users are on one side and the physicians are on the other side, with a service center, where the data is stored, in between. Only the users and the physicians can access the data stored in the central service center. This prevents users' statistics from being leaked or misused [4,5,7]. It also has the function of identifying persons having signs of upcoming severe events.

In conclusion, Ambient Assisted Living (AAL) systems have the great potential to improve elderly's independency in living, but there are some noteworthy issues and factors in AAL designing. Three methods mentioned in the article could satisfy the major aim of AAL and resolve issues. Radar system and SafeNeighborhood system are all promising methods which can alleviate the manual effort and secure users' data safety. However, both of two methods have technical shortcomings. Thus, it is recommended that our company focus on the final method IOT. It is proved to be technically sophisticated. By applying it, AAL could assist elderly to access information effortlessly and receive timely medical help without worrying about other issues.

[1050words]

Glossary：

IOT: the abbreviation of "The Internet of Things", which refers to anything connected to the Internet [8].

NFC: the abbreviation of "Near Field Communication", which is a short-range wireless technology that enables simple and secure communication between electronic devices [9].

References:

[1] C. Debes, A. Merentitis, S. Sukhanov, M. Niessen, N. Frangiadakis, and A. Bauer, "Monitoring Activities of Daily Living in Smart Homes: Understanding human behavior," IEEE Signal Processing Magazine, vol. 33, no. 2, pp. 81-94, Mar. 2016. doi: 10.1109/MSP.2015.2503881.

[2] A. C. Bicharra Garcia, A. S. Vivacqua, N. Sánchez-Pi, L. Martí and J. M. Molina, "Crowd-Based Ambient Assisted Living to Monitor the Elderly's Health Outdoors," IEEE Software, vol. 34, no. 6, pp. 53-57, November/December 2017, doi: 10.1109/MS.2017.4121217.

[3] V. Rogelj and D. Bogataj, "Ambient Assisted Living Technologies and Environments: Literature review and research agenda," in 2020 7th International Conference on Control, Decision and Information Technologies (CoDIT), Prague, Czech Republic, 2020, pp. 762-767, doi: 10.1109/CoDIT49905.2020.9263932.

[4] C. Maan and U. Gunawardana, "Barriers in acceptance of Ambient Assisted Living technologies among older Australians," in 2017 IEEE Life Sciences Conference (LSC), Sydney, NSW, Australia, 2017, pp. 222-225, doi: 10.1109/LSC.2017.8268183.

[5] F. Fioranelli, J. Le Kernec and S. A. Shah, "Radar for Health Care: Recognizing Human Activities and Monitoring Vital Signs," IEEE Potentials, vol. 38, no. 4, pp. 16-23, July-Aug. 2019, doi: 10.1109/MPOT.2019.2906977.

[6] P. Rashidi and A. Mihailidis, "A Survey on Ambient-Assisted Living Tools for Older Adults," IEEE Journal of Biomedical and Health Informatics, vol. 17, no. 3, pp. 579-590, May 2013, doi: 10.1109/JBHI.2012.2234129.

[7] A. Dohr, R. Modre-Opsrian, M. Drobics, D. Hayn and G. Schreier, "The Internet of Things for Ambient Assisted Living," in 2010 Seventh International Conference on Information Technology: New Generations, Las Vegas, NV, USA, 2010, pp. 804-809, doi: 10.1109/ITNG.2010.104.

[8] Christensson, Per. "Internet of Things Definition." TechTerms.com. (January 16,2015).Accessed:May.18,2024.[Online]Available:https://techterms.com/definition/internet_of_things.

[9] Christensson, Per. "NFC Definition." TechTerms.com. (April 7, 2018). Accessed : May 18, 2024. Available:https://techterms.com/definition/nfc.